Zain Ul Abidin

Software Engineer

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Summary

Full-stack software engineer dedicated to building scalable, user-centric applications through clean code, modern system design, and efficient deployment pipelines.

Education

The University of Chakwal — Chakwal, Pakistan
Bachelor of Science in Computer Science — *2022 – 2026*

- **CGPA:** 3.43 / 4.00
- **Relevant Coursework:** Data Structures and Algorithms, Object-Oriented Programming, Database Management Systems, Web Engineering, Operating Systems, Software Engineering, Artificial Intelligence

Technical Skills

Languages: JavaScript, TypeScript, Solidity, Python, C++
Frontend: HTML, CSS, React, Next.js, Tailwind CSS,
Backend: Node.js, Express.js, RESTful APIs, GraphQL, Socket.io
Databases: MongoDB, PostgreSQL, Redis
DevOps & Tools: Git, GitHub, VS Code, Vercel, Render, Docker, Linux, Postman

Projects

Picket: Agentic Candidate Screening Platform — Jan 2026 – May 2026
picket-hr.vercel.app

- Architected a real-time assessment pipeline with sub-80ms UI latency using React, Node.js, and Socket.io; integrated Redis and BullMQ to handle concurrent processing queues and eliminate database bottlenecks.
- Filtered 85% of recruiting pipeline noise by orchestrating a multi-agent AI verification framework (Gemini 2.5 Flash, Llama 3.3, Tavily) to cross-examine web footprints and isolate prompt-injection attempts.
- Secured testing integrity with 99.2% anti-bot accuracy via an interactive Proof-of-Work portal that evaluates telemetry data such as keystroke intervals, paste actions, and latency logs.

Attestify: Blockchain-based Academic Credential Infrastructure — Oct 2025 – Mar 2026
attestify-alpha.vercel.app

- Enabled bulk issuance of 500+ student transcripts by building a full-stack React/Express portal with an asynchronous PDFKit generation engine and dynamic QR codes.
- Prevented credential forgery by developing a custom ERC-721 Soulbound Token (SBT) smart contract in Solidity, overriding internal hooks to enforce strictly non-transferable ownership.
- Reduced blockchain operational costs by 99% and achieved <1.5s verification speeds using a hybrid storage architecture (IPFS via Pinata) and anchoring SHA-256 hashes on the Sepolia testnet.
- Eliminated transaction collisions during bulk operations by engineering an asynchronous Node.js Mutex handler coordinated with Solidity batch minting methods.

Publications

Attestify: A Hybrid Blockchain-IPFS Framework for Credential Verification — June 2026
Published in Spectrum of Engineering Sciences

- Authored a paper proposing a novel decentralized architecture combining non-transferable Soulbound Tokens (SBTs) and dual-anchor document integrity (SHA-256 and IPFS) to eliminate academic fraud.
- **DOI:** 10.5281/zenodo.20655578 | **Read Paper:** thesesjournal.com

Certifications

Web Programming with Python and JavaScript (Harvard CS50) — *August 2025*
Certified React Developer (HackerRank) — *July 2025*
Blockchain Basics (Cyfrin Updraft) — *October 2025*